

KeynetikPOS

Session Log

Instructions issued and work performed

Working session of 19 July 2026

Repository: KeynetikPOS | Branch: feature/erp-tier4

Topics: M-Pesa till payments, installer packaging, release versioning

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1 Scope and how to read this document

This document records a single working session held on 19 July 2026 against the **feature/erp-tier4** branch of KeynetikPOS. It captures, in order, each instruction that was issued, the investigation carried out in response, the conclusion reached, and any commands run or files changed.

It is a record of **what was asked and what was done** — not a specification and not a substitute for the technical documentation. Where a conclusion rests on something that was verified by reading code, the relevant file and line is cited. Where a conclusion rests on external behaviour that was *not* verified in this session — principally the behaviour of Safaricom's Daraja API — that is stated explicitly rather than presented as established fact.

Verification status

Every claim about this codebase in this document was checked against the source during the session. Claims about Daraja's Paybill/Buy Goods semantics come from API knowledge and were **not** tested against a live or sandbox Daraja endpoint. They are flagged in section 4 and section 11.

Session at a glance

Item	Outcome
Requests handled	7
Source files changed	3 (index.js, README.md, CMakeLists.txt)
Installers produced	2 (v2.0.0, then v2.1.0)
Test suites run	7 of 7 passing, twice
Commits made	None — all changes left in the working tree
Deployments made	None — the Worker change is not live

Nothing was committed or deployed

The session ended with all changes uncommitted in the working tree and the Cloudflare Worker still running the previous code. Both were deliberate — see section 11.

2 Request 1 — Is M-Pesa till payment wired up?

Instruction: “was this wired up to work with mpesa till payments?”

2.1 Investigation

A repository-wide search for M-Pesa, till, paybill, Daraja and STK-push terms returned 48 files. The three that carry the implementation were read in full.

- **mpesaclient.h / mpesaclient.cpp** — the desktop client.
- **license-server/src/index.js** — the Cloudflare Worker endpoints.
- **paymentdialog.cpp** — the cashier-facing wiring.

2.2 What was found

The feature is implemented end to end. The architecture routes every push through the licence Worker rather than talking to Daraja from the till.

Layer	Implementation
Client	MpesaClient::requestPayment() POSTs to /pos/mpesa/stkpush , authenticated with the till's licence key and device id, then polls /pos/mpesa/status every 3 s for up to 25 attempts (75 s). Results surface as the signals promptSent , paymentConfirmed(receipt) and paymentFailed(reason) .
Worker	Initiates the Daraja STK push, receives Safaricom's callback at /pos/mpesa/callback , and serves the outcome from the stk_requests table in D1. Daraja credentials are Worker secrets and never reach the shop PC.
Cashier UI	A Mobile Money payment line with a <i>Send STK Prompt</i> button; on confirmation the receipt code is written into the reference field, validated against ^[A-Z0-9]{10}\$. Manual entry remains available when the till is unlicensed or the push times out.

The reason for the Worker sitting in the middle is documented at **mpesaclient.h:13** — a desktop till behind NAT cannot receive Safaricom's callback, so a public endpoint is required as the rendezvous point.

2.3 The defect identified

Buy Goods (till) sends the wrong PartyB

At the time of review, **license-server/src/index.js** sent **BusinessShortCode** and **PartyB** as the same value (**MPESA_SHORTCODE**). That is correct for Paybill. For Buy Goods, Daraja expects **BusinessShortCode** to be the Head Office / store number — which is also what the **Password** hash is built from — and **PartyB** to be the till number the customer keys in. With a single value serving both, a genuine Buy Goods configuration would fail. Fixed later in the session; see section 8.

A secondary observation: the C2B handler at **/webhook/mpesa** is for licence purchases and assigns licence keys. It is unrelated to POS sales, which travel exclusively through the STK path.

3 Request 2 — Clarification: for POS sales

Instruction: “i meant for POS sales”

The clarification did not change the answer: the STK path **is** the POS sale path. The sale-side persistence was then traced to confirm the loop closes.

3.1 How a confirmed payment reaches the sale

- 1 **getTenders()** (paymentdialog.cpp:447) appends { **"Mobile Money", amount, reference** }, where the reference is the M-Pesa receipt code written into the field by the **paymentConfirmed** handler.
- 2 **getPaymentMethod()** reports *Mobile Money*, or a joined string such as *Cash + Mobile Money* for split tenders.
- 3 **getReferenceNumber()** prefixes the code with *M-Pesa* for printing on the receipt.
- 4 Confirm stays disabled until a valid ten-character code is present, so a sale cannot close against an unconfirmed push.

Conclusion

Cashier tenders → prompt to the customer's phone → code returns → sale records it. The loop is complete and manual fallback works.

4 Request 3 — A test till number for Buy Goods

Instruction: “what fake till can i use to test buy goods till function on this POS?”

4.1 Guidance given

No till number was invented. Sandbox shortcodes are issued per Daraja account, so a fabricated number would be rejected at Daraja rather than exercising the POS code path — the failure would teach nothing.

- **Where the real values live:** the Daraja portal, under the application's *Test Credentials / Simulator* tab. These are tied to the account and are not transferable.
- **Universally available in sandbox:** the M-Pesa Express pair — shortcode **174379** with the public sandbox passkey, and the test MSISDN **254708374149**. Sandbox STK is only really wired for that shortcode.

4.2 The limitation that matters

Sandbox cannot validate a Buy Goods pairing

Sandbox does not meaningfully enforce the relationship between the Head Office number and the till number. A buygoods configuration will return **ResponseCode: 0** and still be wrong in production. Sandbox therefore proves the request shape, callback handling, polling and tender recording — but **not** the till pairing itself, which can only be confirmed by a small live transaction at go-live.

Not verified in this session

The Paybill-versus-Buy-Goods field semantics described here and in section 8 were not tested against a live or sandbox Daraja endpoint. They should be confirmed against current Daraja documentation before go-live.

5 Request 4 — Cloudflare or Supabase for the payment flow

Instruction: a diagram of the STK push flow, with “how would we implement these for POS sales using cloudflare or should we go the supabase way”

5.1 Finding: the design was already built

The supplied diagram described the existing implementation rather than missing work. Each element maps onto shipped code:

Diagram element	Existing code
Qt POS → Backend → M-Pesa → phone	requestPayment() → /pos/mpesa/stkpush → Daraja (mpesaclient.cpp:84)
M-Pesa → Backend (callback)	/pos/mpesa/callback writes status and receipt (index.js:423)
Qt polls /status	poll() → /pos/mpesa/status (mpesaclient.cpp:136)
Every 2–5 seconds	kPollMs = 3000, 25 polls, 75 s ceiling
Backend updates status	stk_requests table in D1 (schema.sql:38)
POS knows → receipt prints	paymentConfirmed → tender plus receipt code

The diagram's central point — that M-Pesa replies to the backend and not to the Qt application — is the documented rationale for the Worker's existence at mpesaclient.h:13.

5.2 Recommendation: remain on Cloudflare

Migrating to Supabase was advised against, for reasons beyond sunk cost:

- **Authentication already exists.** The Worker is the licence server, so **posAuthorised()** reuses the same licence-key and device-id pair as **/validate**. M-Pesa authentication came for free; a migration would mean reimplementing it against a different identity store.
- **Polling is the correct choice here**, not a limitation. The worst case is three seconds of latency on a transaction where the customer is physically present entering a PIN, which takes 15–30 s. A WebSocket buys nothing a cashier can perceive while adding a connection that breaks on unreliable shop connectivity. Polling is stateless and self-healing.
- **Workers is the better webhook receiver** — no cold start, and Safaricom's retry behaviour is unforgiving of latency.
- **Realtime is not exclusive to Supabase.** Durable Objects plus WebSocket or SSE provide the same capability if it is ever wanted.

Gap noted in passing — unreconciled payments

If a callback arrives while the till is offline or closed, the sale is lost from the cashier's view even though the customer was charged. The receipt exists in **stk_requests**, but nothing reconciles it. A “look up an unclaimed M-Pesa payment by phone and amount” screen would close this. Recommended before go-live; not built in this session.

6 Request 5 — How to build the installer

Instruction: “how do i buil installer for this?”

6.1 Finding: packaging was already configured

CPack with the NSIS generator was already wired into CMakeLists.txt, and a prior installer existed in the release build directory. All prerequisites were present: NSIS at **C:\Program Files (x86)\NSIS**, **LICENSE.txt**, and **resources/app.ico**.

```
$env:PATH = "C:\Qt\Tools\mingw1310_64\bin;C:\Qt\6.11.0\mingw_64\bin;" + $env:PATH
cmake --build "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release"
cpack --config "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release\CPackConfig.cmake" \
-B "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release"
```

6.2 How the packaging works

A POST_BUILD step (CMakeLists.txt:164) runs **windeployqt** into **windeployqt_stage/**, collecting the Qt DLLs, platform plugins and SQL drivers beside a copy of the executable. **install()** ships that directory and CPack's NSIS generator wraps it, adding Start Menu and desktop shortcuts, matching uninstaller entries, the application icon and a run-on-finish option.

6.3 Two problems flagged

Problem	Detail
build-release.bat is stale	It configures a different build directory, runs windeployqt in place rather than into the staging directory, and stops before cpack — so it never produces an installer. It also deletes the build tree first. Recommended: fix it to call cpack against the Qt Creator directory, or delete it.
Version was hardcoded	CPACK_PACKAGE_FILE_NAME derives from APP_VERSION, so shipping without bumping it silently overwrites the previous installer with an identically named file.

7 Request 6 — Build the installer

Instruction: “yup”

This reply was ambiguous between two offers — building the installer, and committing the branch work first. It was read as the former, and the commit was deliberately left alone as the more consequential and less reversible of the two.

- 1 Build ran and compiled nothing — the tree was already current.
- 2 All seven ctest suites passed: carttotals, database, ledger, payroll, vat, salejournal, checkoutservice.
- 3 CPack produced **KeynetikPOS-2.0.0-Setup.exe**, 28.2 MB.

Two caveats recorded at the time

The build overwrote the previous 2.0.0 installer, since the filename was unchanged. And the installer contained the then-unfixed M-Pesa till defect — acceptable for Paybill or sandbox use, not for a Buy Goods merchant.

8 Request 7 — Till-number fix and version bump

Instruction: “both”

8.1 The till-number split

license-server/src/index.js:395 — **PartyB** now resolves independently of **BusinessShortCode**:

```
const buyGoods = env.MPESA_TXN_TYPE === "buygoods";
const txnType = buyGoods ? "CustomerBuyGoodsOnline" : "CustomerPayBillOnline";
const partyB = (buyGoods && env.MPESA_TILL_NUMBER)
  ? String(env.MPESA_TILL_NUMBER) : env.MPESA_SHORTCODE;
```

BusinessShortCode and the **Password** hash remain on **MPESA_SHORTCODE**. The fallback means an unset **MPESA_TILL_NUMBER** reproduces the previous request byte for byte, so Paybill behaviour is unchanged.

Transaction type	BusinessShortCode	PartyB
Paybill	Paybill number	The same number
Buy Goods	Head Office / store number	The till number

The same table was added to **license-server/README.md:148** along with the sandbox warning, and the environment-variable header in `index.js` was expanded to document **MPESA_TILL_NUMBER**. **node --check** passed on the modified Worker source.

8.2 The version bump

2.0.0 → **2.1.0** in both **project()** (**CMakeLists.txt:2**) and **APP_VERSION** (**CMakeLists.txt:21**). Changing the **project()** version forces a full CMake reconfigure and recompile, so this rebuild was not incremental.

8.3 Result

Artefact	Size	Built
KeynetikPOS-2.0.0-Setup.exe	28.2 MB	19:05
KeynetikPOS-2.1.0-Setup.exe	28.2 MB	19:57

Both installers now coexist — the version bump achieved its purpose. The full rebuild completed cleanly and all seven test suites passed against it.

9 Consolidated command reference

Every command that materially advanced the work, in the order run. All were executed from the repository root in PowerShell.

Build the release tree

```
$env:PATH = "C:\Qt\Tools\mingw1310_64\bin;C:\Qt\6.11.0\mingw_64\bin;" + $env:PATH  
cmake --build "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release"
```

Run the test suite

```
ctest --test-dir "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release" --output-on-failure
```

Produce the installer

```
$env:PATH = "C:\Program Files (x86)\NSIS;" + $env:PATH  
cpack --config "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release\CPackConfig.cmake" \  
-B "build\Desktop_Qt_6_11_0_MinGW_64_bit-Release"
```

Syntax-check the Worker

```
node --check "license-server\src\index.js"
```

Deploy the Worker — NOT run in this session

```
cd license-server  
wrangler deploy --var MPESA_TILL_NUMBER:<till>  
# or set MPESA_TILL_NUMBER in wrangler.toml [vars], then: wrangler deploy
```

The deploy command above was not executed

It is recorded for completeness. The live Worker still runs the previous code, and deployment was deliberately deferred until a real till number is available.

10 Changes made to the repository

Three files were modified. None were committed.

File	Change
license-server/src/index.js	PartyB reads MPESA_TILL_NUMBER for buygoods, falling back to MPESA_SHORTCODE. BusinessShortCode and Password unchanged. Environment-variable header documents the new variable and the HO-versus-till distinction.
license-server/README.md	Added the Paybill / Buy Goods field table, the MPESA_TILL_NUMBER setup line, and the warning that sandbox does not validate the pairing.
CMakeLists.txt	APP_VERSION and project() version raised from 2.0.0 to 2.1.0.

Pre-existing uncommitted work

The branch already carried roughly twenty modified files and one untracked test file (tests/checkoutservice_test.cpp) before this session began. Those changes are unrelated to this work but are included in the 2.1.0 installer, since it was built from the working tree. Splitting them into separate commits was offered and not taken up.

11 Outstanding items

In the order they matter.

#	Item	Detail
1	Worker not deployed	Live traffic still reaches the previous code. Requires wrangler deploy with MPESA_TILL_NUMBER set — worth doing only once the real till number is in hand.
2	Nothing committed	No commit reproduces the 2.1.0 binary. The branch holds this session's three files plus the pre-existing changes.
3	Buy Goods pairing untested	Sandbox returns success regardless of the pairing, so only a small live transaction confirms it.
4	Clean-machine install test	windeployqt occasionally misses a plugin that surfaces only at runtime on a machine without Qt installed; the SQL driver is the usual suspect.
5	Unclaimed-payment reconciliation	No way to recover a payment whose callback landed while the till was offline. Recommended before go-live (section 5.2).
6	build-release.bat is misleading	Does not produce an installer and targets the wrong build directory. Fix or delete (section 6.3).

Record of the working session of 19 July 2026. Generated from the session transcript by docs/generate_session_log.py.